

RAFFLES INSTITUTION

2014 Year 6 Preliminary Examination
Higher 1

BIOLOGY

8875/01

Paper 1 Multiple Choice

29th September 2014

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C, and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

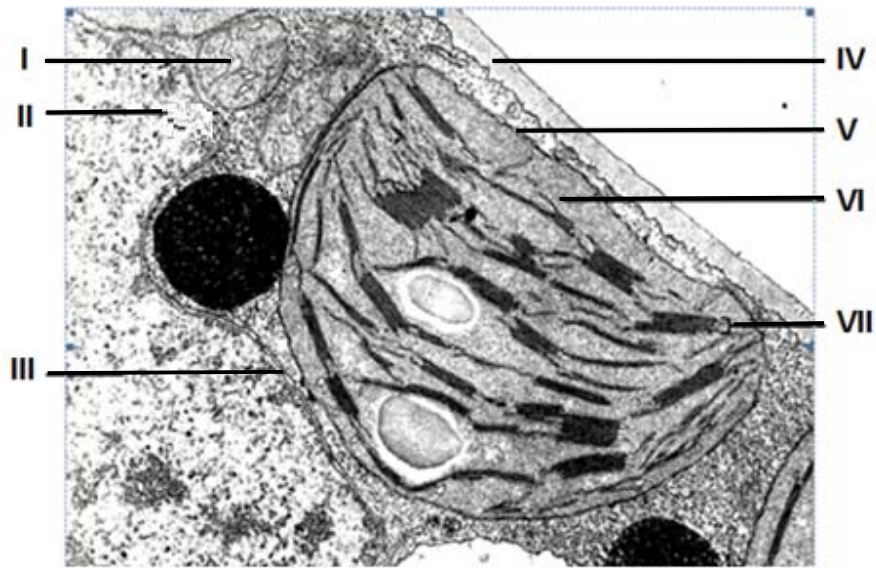
(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

This document consists of **20** printed pages.



Raffles Institution
Internal Examination

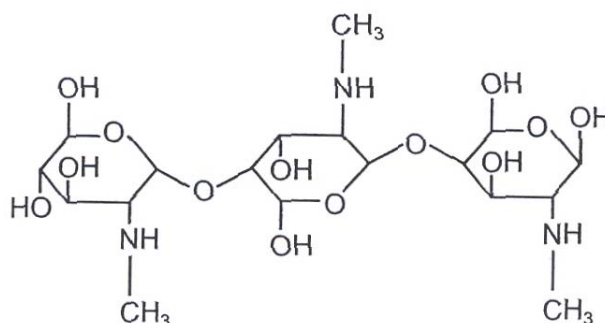
1. The diagram below shows part of a eukaryotic cell.



Which of the structures labelled I to VII are correctly matched to the structures/processes in the table below?

	composed of two membranes	site of Calvin cycle	composed of cellulose	stack of membrane bound structures	site of ATP synthesis
A	III	VI	IV	VII	I
B	III	IV	VI	V	I
C	V	VI	IV	VII	II
D	V	VI	III	I	II

2. The diagram shows the structure of a competitive inhibitor of the enzyme lysozyme.



Which substance is most likely to be the normal substrate of lysozyme?

- A Lipid
 B Polynucleotide
 C Polypeptide
 D polysaccharide
3. A student investigating the relative carbohydrate concentrations of different brands of strawberry jam obtained the following results.

Which brand of jam contained the lowest concentration of sucrose?

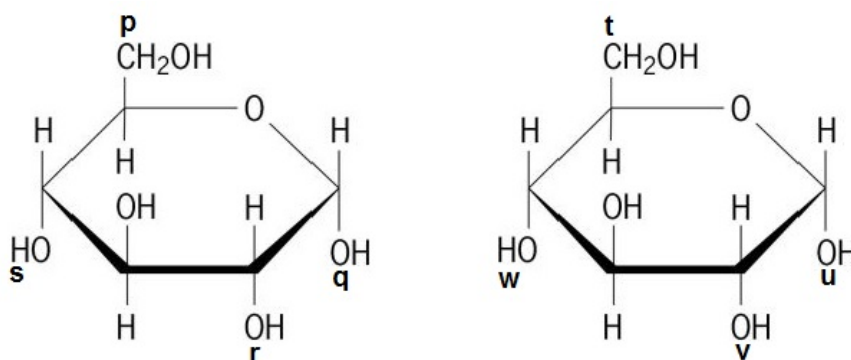
brand of jam	total reducing sugar content per 100g of jam (arbitrary units)	
	untreated jam	jam after hydrolysis with invertase
A	54	76
B	60	84
C	73	93
D	43	65

4. The table describes some molecules found in a cell surface membrane.

Which of the following is correct?

	cholesterol	glycolipid	glycoprotein	phospholipid	protein
A	is a precursor of certain hormones	involved in active transport	transports substances across the membrane	is amphipathic in nature	involved in DNA replication
B	regulates membrane fluidity at different temperatures	involved in cell-cell recognition	increases membrane fluidity at high temperature	involved in DNA replication	limits the permeability of the membrane
C	decreases membrane fluidity at high temperature	involved in cell-cell recognition	acts as a signal binding site during cell signaling	limits the permeability of the membrane	can change in conformation when phosphorylated by ATP
D	has a four ring structure	involved in active transport	acts as a ligand binding site	is amphoteric in nature	limits the permeability of the membrane

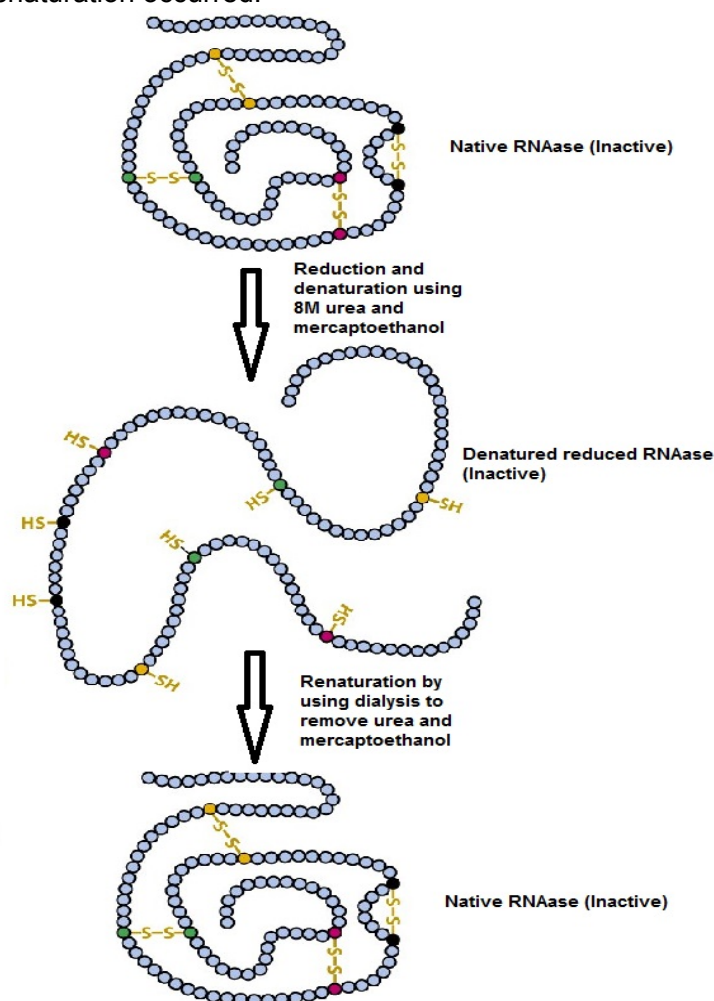
5. The diagram shows two molecules of glucose. Four possible bonding positions are labelled p, q, r and s on one molecule and t, u, v and w on another molecule.



Which of the following bonding positions are involved in the formation of glycogen?

- A** p-u and p-v
- B** p-u and q-w
- C** p-v and q-w
- D** p-w and v-w

6. An experiment was carried out using pancreatic RNAase. A reducing agent, mercaptoethanol, and urea were added and denaturation was observed. Upon removal of the urea and mercaptoethanol, renaturation occurred.

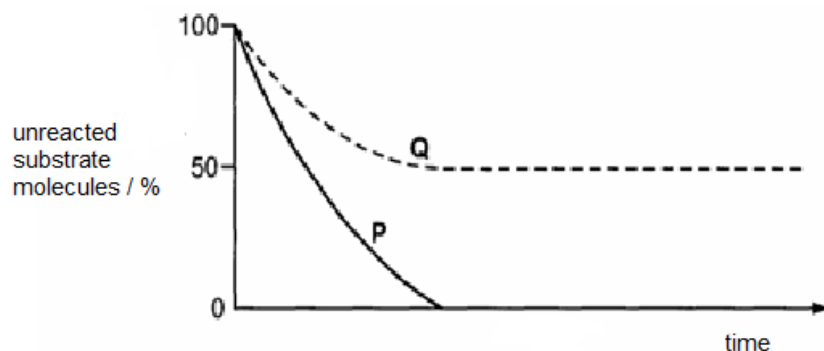


Based on the above experiment, which of the following statements are true?

- I The polypeptide chain must have refolded as directed by its primary sequence and once refolded, the original disulphide bonds must have formed again.
 - II Given the right conditions, denatured proteins can refold.
 - III The tertiary structure of a protein is not determined by its primary structure.
 - IV Denaturation alters the primary structure of a protein and destroys its biological activity.
- A I and II only
- B II and IV only
- C II and III only
- D I and IV only

- 7 Curve **P** represents the course of an enzyme-catalysed reaction under optimum conditions. A single factor was then changed in the reaction mixture.

Curve **Q** shows the effect the reaction rate of enzyme **P** after this change.



Which factor, operating at a constant extent throughout the experiment, could show the result shown by curve **Q**?

- A a higher pH
 - B a lower temperature
 - C an end-product inhibitor
 - D an increase in substrate concentration
8. Equal quantities of a 5% starch solution were added to equal quantities of different solutions. After 5 minutes at room temperature, the solutions were tested with iodine in potassium iodide solution and the depth of colour was recorded using a colorimeter (with an arbitrary scale of 1 to 5, 5 being the darkest blue). The results are given in the table.

solution	colorimeter reading
saliva and dilute HCl	4
saliva and dilute NaHCO ₃	1
Dilute HCl only	4
Saliva at 100°C	5
Saliva at 20°C	3

This experiment suggests that saliva contains a biologically active substance that

- A hydrolyses starch in acid conditions.
- B is denatured by hydrochloric acid.
- C is promoted in its action by high temperatures.
- D may be activated by hydrogencarbonate ions.

9. The figures below show the complete karyotypes of 2 rodents of the same species. In this species of rodent, males are the heterogametic sex.



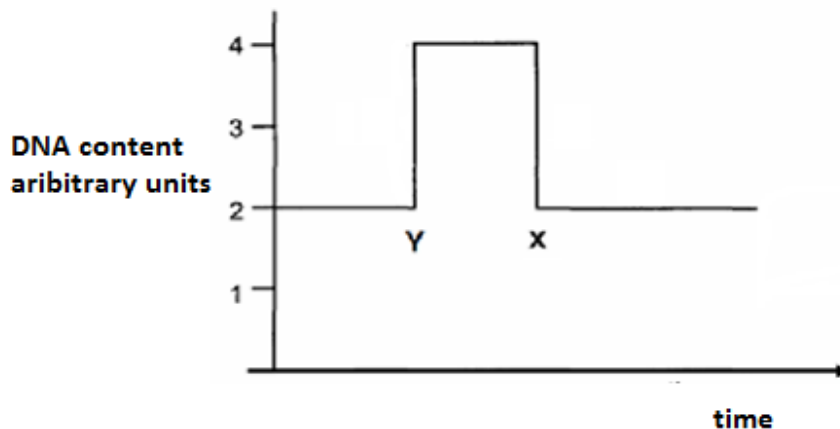
Rodent A (Male)



Rodent B (Female)

Which of the following observations is not true?

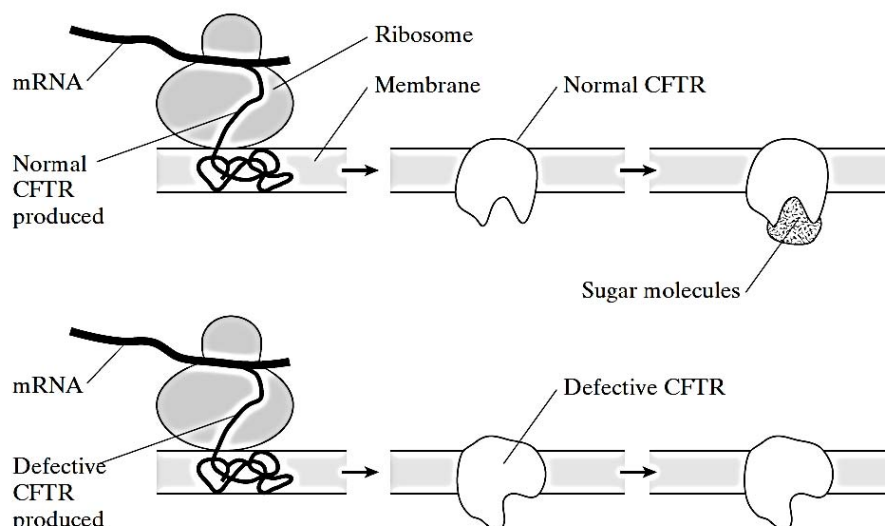
- A A chromosomal aberration occurred in the ovary of the mother of rodent B.
 - B Rodent A is diploid and $2n = 16$.
 - C Rodent B has 1 missing X chromosome.
 - D Non-disjunction of autosomes occurred in rodent B.
10. The graph shows the amount of DNA present in the nuclei of cells undergoing division in a multicellular organism.



Which combination is correct?

	Y	X
A	anaphase	interphase
B	interphase	telophase
C	telophase	anaphase
D	prophase	telophase

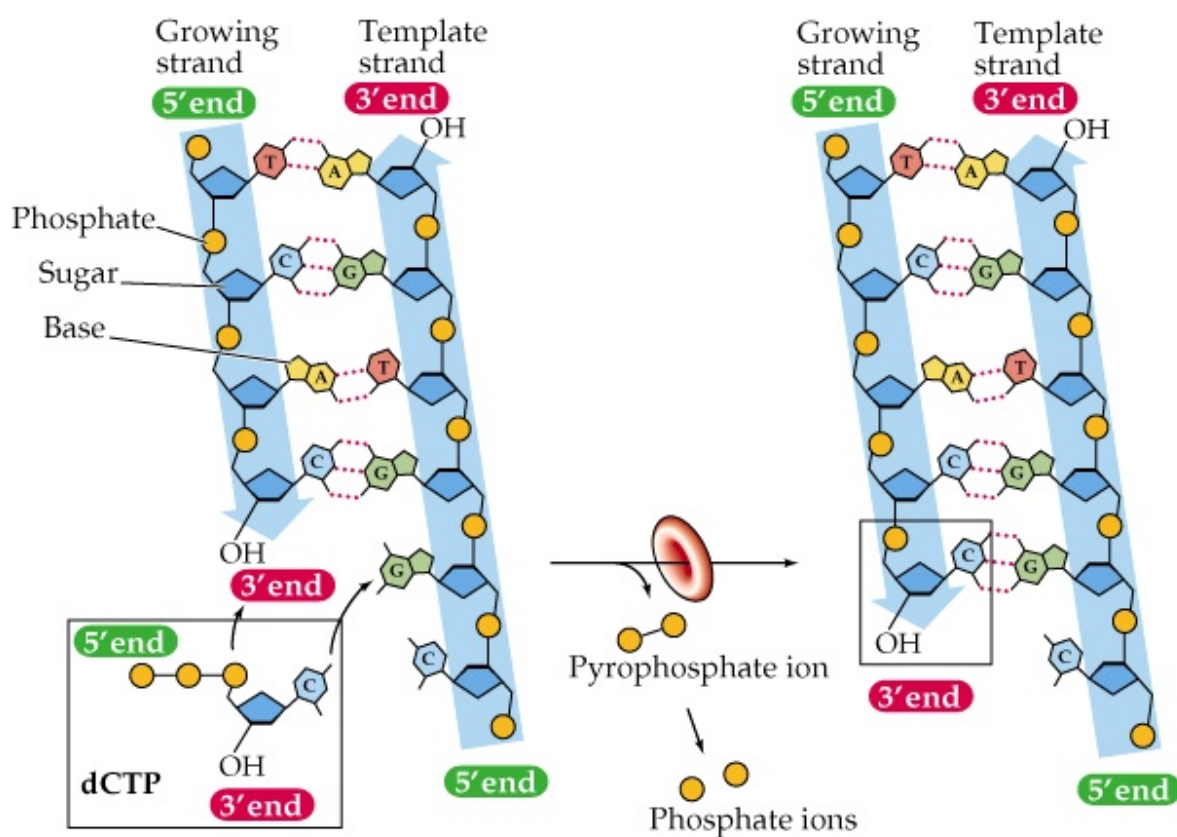
11. CFTR is a transmembrane regulator protein. It is made up of 1480 amino acids. People with cystic fibrosis produce a defective CFTR protein which is missing one amino acid from its structure. The diagram shows the synthesis of a normal and a defective CFTR in a cell. A normal CFTR protein molecule has sugar molecules attached to it which make it functional.



Which of the following statements are true?

- I 4440 is the number of bases on the template DNA which code for the amino acid sequence of a normal CFTR protein.
 - II Cystic fibrosis is due to a chromosomal mutation known as a deletion.
 - III The functional CFTR is a protein that undergone glycosylation.
 - IV The defective CFTR is not functional as the mutation in the DNA resulted in changes in its tertiary structure.
- A I and IV only
- B I and III only
- C I, III and IV only
- D II, III and IV only

12. During DNA replication, nucleotides are added to the 3' end of a growing polynucleotide.



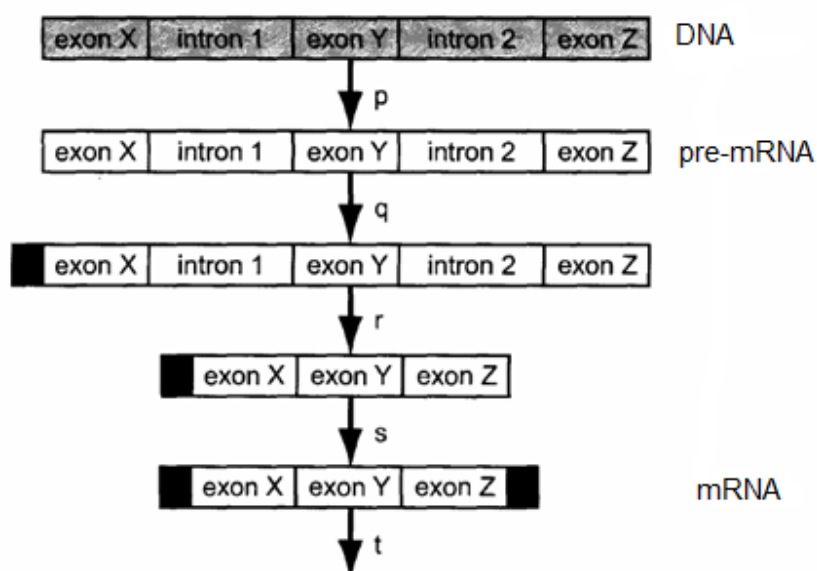
Which of the following correctly matches type of bond(s) that is/are formed with the processes that are involved in elongation of the growing DNA strand?

	type of bond(s) formed	processes involved
A	phosphodiester and hydrogen	hydrolysis, polymerisation and complementary base pairing
B	phosphodiester	hydrolysis and condensation
C	phosphodiester and hydrogen	hydrolysis, condensation and complementary base pairing
D	hydrogen	complementary base pairing

13. Cytosine comprises 32% of the nitrogenous bases in a sample of DNA consisting of 200 base pairs.
What is the total number of adenine and thymine bases in a single strand of DNA from this sample?

A 16
B 18
C 36
D 72

14. The diagram represents the processing of pre-messenger RNA.



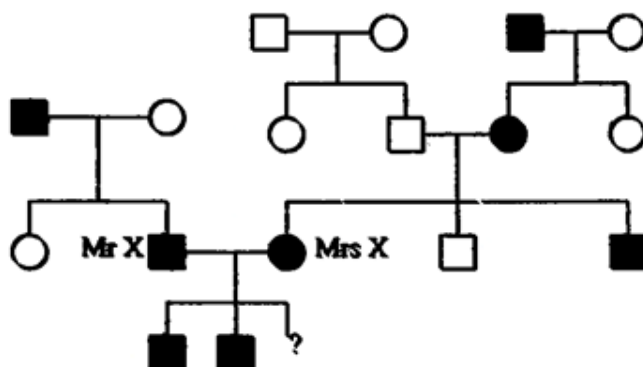
During the process of pre-messenger RNA each of the listed events occurs:

- 1 capping
- 2 polyadenylation
- 3 splicing
- 4 transcription
- 5 translation

Which correctly identifies the processes **p**, **q**, **r**, **s** and **t**?

	p	q	r	s	t
A	5	1	3	2	4
B	4	1	3	2	5
C	5	2	1	3	4
D	4	2	1	3	5

15. The family tree below was assembled by a genetic counselor for Mr and Mrs X who suffer from heart disease due to hypercholesterolaemia. Children who inherit the dominant mutant allele from both parents rarely survive beyond puberty.



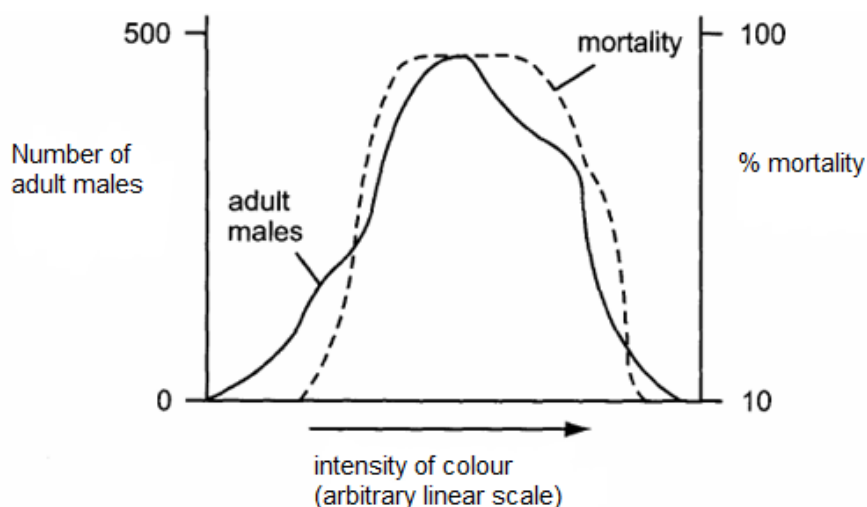
Key

affected male	■
unaffected male	□
affected female	●
unaffected female	○

What is the probability that Mr and Mrs X's third child will be unaffected?

- A 0.75
 B 0.50
 C 0.25
 D 0.00

16. The graph was drawn from data on a population of a species of moth which shows considerable variation in colour intensity.



Which conclusion can be made from this graph?

- A Colour variation is environmentally induced.
 - B Colour variation is genetically determined.
 - C Extreme forms are favoured by selection.
 - D The species shows discontinuous variation with respect to colour.
17. A population of cats contains individuals with shortened tails known as Manx cats and individuals with normal tails known as non-Manx cats. Mating of two non-Manx cats always produced non-Manx cats. Mating of two heterozygous Manx cats always produced a mixture of Manx cats and non-Manx cats, where the number of Manx cats is usually twice that of non-Manx cats.

Which hypothesis is consistent with these observations?

- A The genes for shortened tails and normal tails are linked on the same chromosome.
- B The genes for shortened tails and normal tails are sex-linked.
- C The allele for shortened tails is dominant over the allele for normal tails and the homozygous dominant genotype results in death.
- D The allele for shortened tails is dominant over the allele for normal tails and the homozygous recessive genotype results in death.

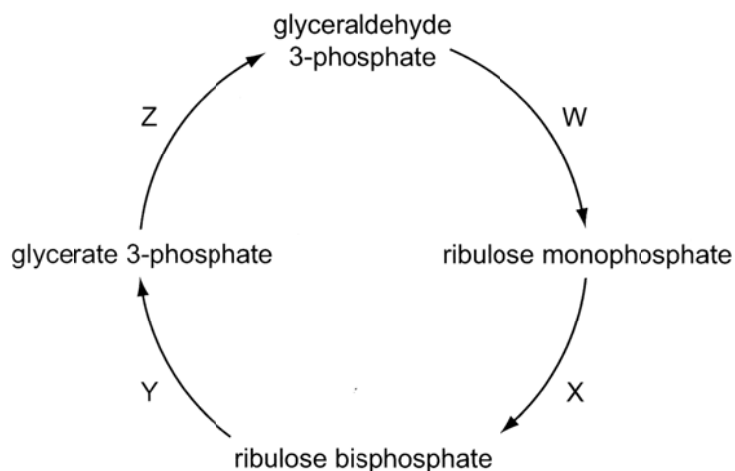
18. In lentils, the seed coat pattern is determined by a gene with 3 alleles, C^M , C^S and C^C , whose phenotypes are marbled, spotted and clear respectively. Four crosses were repeated many times. The crosses and the outcomes of these crosses are shown in the table below.

cross	parents	offspring phenotype and ratio
1	marbled x marbled	3 marbled : 1 clear
2	spotted x clear	all spotted
3	marbled x marbled	3 marbled : 1 spotted
4	clear x clear	all clear

From the data, it is possible to conclude that,

- A spotted is recessive to clear.
- B all of the clear offspring are heterozygous.
- C two thirds of the marbled offspring in cross 3 are heterozygous.
- D the marbled parents in cross 1 have the same genotype as the marbled parents in cross 3.

19. The diagram shows the main stages in the Calvin cycle.



At which stages are ATP and reduced NADP used and carbon dioxide taken up?

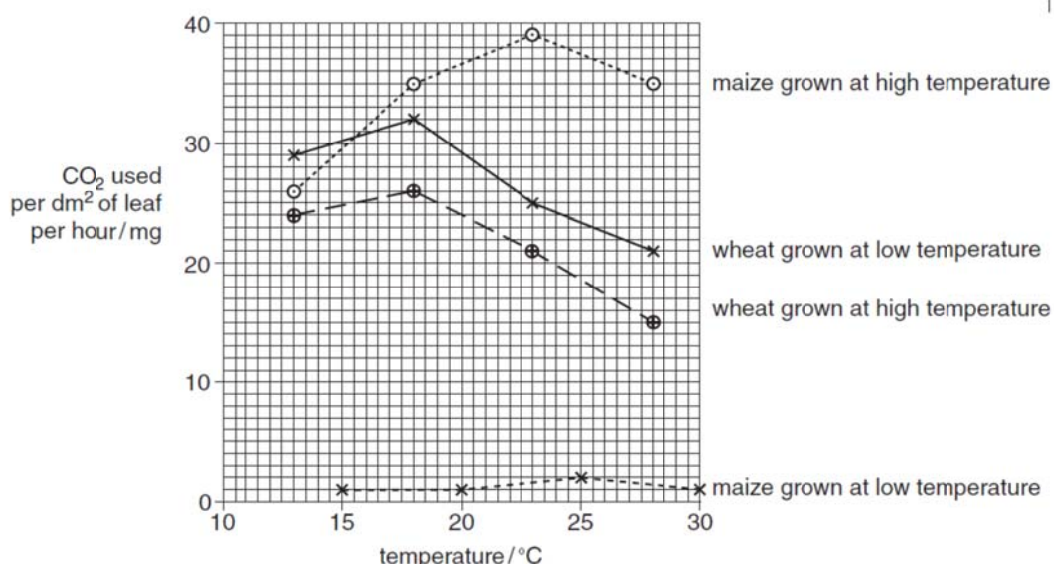
	ATP used	reduced NADP used	carbon dioxide taken up
A	W and Z	X	Y
B	X	W and Y	Z
C	Y	X and Z	W
D	X and Z	Z	Y

20. During electron transport, each reduced NAD produces on average 2.5 molecules of ATP and each reduced FAD produces on average 1.5 molecules of ATP.

Which of the following shows the correct number of ATP molecules produced in each stage of cellular respiration, from the oxidation of 1 molecule of glucose?

	molecules of ATP produced in each stage of cellular respiration			
	glycolysis	link reaction	Krebs cycle	electron transport
A	1	0	1	14
B	2	0	2	18
C	2	0	2	28
D	2	1	2	28

21. Young maize and wheat plants were grown to maturity at high and low temperatures. The rate of photosynthesis in each of these mature plants was measured at different temperatures. The rate of photosynthesis was measured as the amount of CO_2 used per dm^3 of leaf per hour. The results are shown in the graph below.



What information can be concluded from the graph above?

- I For plants grown at high temperature, the rate of photosynthesis is optimum at 25°C in maize and 18°C in wheat.
 - II For plants grown at high temperature, maize had a greater increase in rate of photosynthesis compared to wheat until optimum temperature was reached.
 - III The rate of photosynthesis was affected more significantly in maize plants than in wheat plants grown at low temperatures.
 - IV Low temperatures slowed down the formation of membranes in maize plants but not in wheat plants which caused a decrease in lamellae formation.
- A II and III only
- B I and IV only
- C I, II and III only
- D All of the above

22. An inhibitor of the enzyme catalyzing the oxidative decarboxylation of α -ketoglutarate (5C) was added to respiring animal tissues.

What would have been the effect of the inhibitor on the concentration of oxaloacetate, pyruvate and ATP?

	oxaloacetate	pyruvate	ATP
A	decreases	increases	decreases
B	decreases	decreases	increases
C	increases	increases	decreases
D	increases	decreases	increases

23. The following describes certain structures found in two organisms.

- I fins of a dolphins and fish
- II legs of insects and lizards
- III wing bones of birds and bats
- IV appendix of the humans and cecum of the woolly monkey

Which of the above are homologous structures?

- A IV only
 - B III and IV only
 - C I and III only
 - D I and II only
24. Which of the following statements about natural selection is false?
- A Natural selection will result in a population acquiring new alleles that will confer advantageous phenotypes that will be selected for.
 - B Natural selection occurs even when the environmental conditions remain unchanged.
 - C Natural selection works to eliminate the unfit from the population through selection pressure, leaving the fit to reproduce.
 - D Natural selection works when the population is under a constant struggle for survival.

25. Listed below are some key concepts in the process of speciation.

- I natural selection
- II changes in allele frequencies
- III accumulation of unique mutations in sub-populations
- IV isolation mechanism
- V disruption of gene flow between populations
- VI reproductive isolation

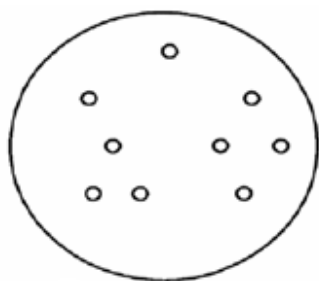
Which of the following shows the correct sequence of events that leads to speciation?

- A VI→V→IV→I→II→III
- B VI→IV→V→III→I→II
- C IV→V→VI→I→II→III
- D IV→V→I→II→III→VI

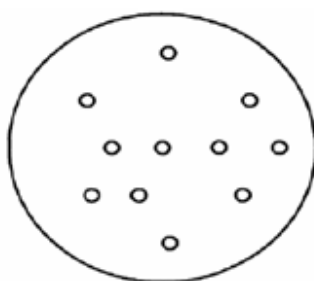
26. Which one of the following is not true about genomic and cDNA libraries?

	genomic library	cDNA library
A	contains entire DNA content of an organism	contains entire protein-encoding DNA content of an organism
B	larger number of different clone types	smaller number of different clone types
C	cannot be used to study changes in gene expression levels	can be used to study gene expression at different stages of development
D	to make a genomic library, genetic material can be isolated from any somatic cell	to make a cDNA library, genetic material can only be isolated from a cell that produces the protein of interest

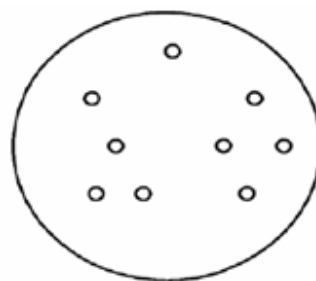
27. A gene coding for the production of a human gene product was inserted into a plasmid with genes coding for resistance to antibiotics ampicillin, streptomycin and tetracycline. The plasmids were used to transform *E. coli* and the bacteria were grown on a nutrient medium with various antibiotics using replica plating. The resulting plates are shown in the diagram.



Medium with ampicillin,
streptomycin and
tetracycline



Medium with ampicillin
and streptomycin



Medium with ampicillin
and tetracycline

Which antibiotic gene(s) contain(s) the restriction site?

- A ampicillin
- B streptomycin
- C tetracycline
- D ampicillin and tetracycline

28. Which one of the following implications of the human genome project is not a concern?
- A Predictive genetic tests become available so that individuals can take steps to manage the disease.
 - B If genetic sequences are patented, it will increase the cost of genetic research and treatment.
 - C New genetic knowledge may conflict with a variety of ethical, social and legal perspectives.
 - D DNA sequences of all individuals are readily accessible to the public.
29. Induced pluripotent stem cells are stem cells that can be generated directly from adult stem cells under the influence of molecular signals.

Which of the following statements are true?

- I An induced pluripotent stem cell can become any cell of the developed organism, but cannot produce trophoblast and placenta to support organismal development, whereas a totipotent stem cell can produce a whole organism including extraembryonic tissue.
 - II A totipotent stem cell and induced pluripotent stem cell can give rise to any cell.
 - III An induced pluripotent stem cell can give rise to a single cell lineage whereas a totipotent stem cell can give rise to multiple, but limited number of cell lineages.
 - IV A totipotent stem cell can become any cell of a developed organism, but cannot produce trophoblast and placenta to support organismal development, whereas an induced pluripotent stem cell can produce a whole organism including extraembryonic tissue.
 - V Induced pluripotent stem cells have the same developmental potential as embryonic stem cells.
- A I only
 - B I and V only
 - C II, III and IV only
 - D III, IV and V only

30. Which of the following is **not** an example of genetically modified organisms?
- A AquaAdvantage Salmon that grow to adult size quickly.
 - B Bt corn plants that are insect-resistant.
 - C Golden Rice that produce high levels of beta-carotene.
 - D milk-producing cows injected with bovine somatotrophin (BST).

End of Paper

Answers:

1.	A	11.	C	21.	A
2.	D	12.	A	22.	
3.	C	13.	C	23.	B
4.	C	14.	B	24.	A
5.	B	15.	C	25.	D
6.	A	16.	C	26.	A
7.	C	17.	C	27.	C
8.	D	18.	C	28.	A
9.	D	19.	D	29.	B
10.	B	20.	C	30.	D

Q16: (N11Q18 post-translation and post-transcription NOT in prok)